

R-VerifyAC Metrics Explained

Cumulative likelihood, confidence, ϵ , and entropy

ANPLOS / MR-BSP working note

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1 The one-sentence interpretation

R-VerifyAC asks:

Across plausible realizations of the missing observations, how often would the joint action selected in Step 1 still be optimal?

That probability is the action’s *cumulative likelihood*. In the continuous-observation implementation it is estimated by Monte Carlo sampling. The variable named **confidence** is the estimated cumulative likelihood of the particular action selected in Step 1.

2 From belief and entropy to a winning action

Let $\bar{a}$ denote the joint action selected from the current belief in Step 1. For every hypothetical missing-observation sample $z^{(i)}$, the algorithm constructs the corresponding hypothetical belief $b^{(i)}$ and evaluates every candidate joint action:

$$a_i^* = \arg \max_{a \in \mathcal{A}} J(b^{(i)}, a).$$

In the present experiments, J is an entropy-based or hybrid objective. A schematic hybrid form is

$$J(b, a) = -\beta H(b, a) - (1 - \beta)L(a),$$

where H is the predicted focused entropy and L is path length. The exact objective determines which action wins each sample. After the winner is identified, however, the magnitude of the entropy is not accumulated into the confidence calculation.

3 Cumulative likelihood

For an action a , define the region of missing-observation space in which that action is optimal:

$$\text{cobs}_a(\mathcal{Z}) = \left\{ z \in \mathcal{Z} : a = \arg \max_{a' \in \mathcal{A}} J(b(z), a') \right\}.$$

The theoretical cumulative likelihood is the probability mass of that region:

$$\text{Cl}_a = \Pr(a^*(Z) = a \mid \mathcal{H}_k^c) = \int_{\mathcal{Z}} p(z \mid \mathcal{H}_k^c) \mathbb{I}[a^*(z) = a] \, dz.$$

“Cumulative” means that the probability mass of *all observations that favor the same action* is combined. It does not mean cumulative over communication iterations, and it does not mean cumulative entropy.

If the possible actions partition the observation space without ambiguous ties, then

$$\sum_{a \in \mathcal{A}} \text{Cl}_a = 1.$$

4 Monte Carlo estimate and the code's confidence

Suppose the algorithm draws N equally weighted samples from the missing-data distribution. Its estimator is

$$\widehat{\text{Cl}}_a = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[a_i^* = a] = \frac{\#\{i : a_i^* = a\}}{N}.$$

For the Step-1 action $\bar{a}$, the implementation stores

$$\text{confidence} = \widehat{\text{Cl}}_{\bar{a}}.$$

Meaning of every symbol

The symbols in the estimator have the following precise meanings:

Symbol	Meaning
$\mathcal{A}$	The finite set of candidate joint actions being compared by the multi-robot planner. An element $a \in \mathcal{A}$ specifies one complete candidate choice for the robot team.
a	One particular candidate joint action whose cumulative likelihood is being calculated. The estimator can be calculated separately for every $a \in \mathcal{A}$.
$\bar{a}$	The particular joint action selected as optimal in Step 1 using the robot's current local belief, before the missing observations are sampled.
N	The total number of Monte Carlo missing-observation samples. Experiments 23 and 24 use $N = 10$ for each Step-2/Step-3 calculation.
i	The index of one Monte Carlo sample, with $i \in \{1, \dots, N\}$.
$z^{(i)}$	The i th hypothetical realization of the observations currently missing from the reasoning robot's belief. It is drawn from the missing-observation distribution implied by the belief and observation model.
$b^{(i)}$	The hypothetical belief obtained after incorporating sample $z^{(i)}$. The planner evaluates all candidate actions under this belief.
$J(b^{(i)}, a)$	The planning objective assigned to candidate action a under hypothetical belief $b^{(i)}$. Entropy influences the estimator through this objective because it can change which action maximizes J .
a_i^*	The winning joint action in sample i : $a_i^* = \arg \max_{a' \in \mathcal{A}} J(b^{(i)}, a')$. The implementation records this winner as <code>opt_action</code> .
$\mathbb{I}[a_i^* = a]$	Indicator function. Its value is 1 if sample i selects action a , and 0 otherwise.
$\sum_{i=1}^N$	Adds the indicator values over all samples. Because every term is either 0 or 1, the sum is exactly the number of samples favoring a .
$\#\{i : a_i^* = a\}$	The cardinality, or count, of sample indices for which action a wins. It is another notation for the preceding sum. Equivalently one may write $ \{i : a_i^* = a\} $.
Cl_a	The true cumulative likelihood of action a : the probability, over the complete missing-observation distribution, that a would be optimal. This is the ideal quantity that would be obtained by integrating over all possible missing observations.
$\widehat{\text{Cl}}_a$	The Monte Carlo estimate of Cl_a . The hat indicates that it is estimated from a finite sample rather than known exactly.

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Symbol	Meaning
$\widehat{\text{Cl}}_a$	The estimated cumulative likelihood specifically for the Step-1 action. In plain language, it is the fraction of sampled missing-data scenarios in which the original Step-1 choice remains optimal.
confidence	The implementation's name for $\widehat{\text{Cl}}_a$. More precisely, the top-level stored value is the reference robot's Step-2 estimate; Step 3 has a separate estimate used in the final Boolean acceptance decision.

For one individual sample, the indicator contribution is therefore

$$\mathbb{I}[a_i^* = a] = \begin{cases} 1, & \text{if action } a \text{ maximizes the objective in sample } i, \\ 0, & \text{otherwise.} \end{cases}$$

For example, if $N = 10$ and action a wins samples $\{1, 2, 4, 6, 8, 10\}$, then

$$\sum_{i=1}^{10} \mathbb{I}[a_i^* = a] = 6, \quad \widehat{\text{Cl}}_a = \frac{6}{10} = 0.6.$$

The estimator assumes that the samples are drawn from the intended missing-observation distribution and are equally weighted. If samples were drawn from a different proposal distribution, the simple count ratio would generally need to be replaced by a weighted importance-sampling estimator.

Therefore, **confidence** is best read as *estimated support probability of the Step-1 action under sampled missing observations*. It is not a covariance confidence interval, localization confidence, mean reward, or entropy value.

Example

With $N = 10$ samples, suppose the winning actions are

$$(3, 3, 1, 3, 0, 3, 1, 3, 0, 3).$$

The estimated cumulative-likelihood distribution is

$$\widehat{\text{Cl}}_0 = 0.2, \quad \widehat{\text{Cl}}_1 = 0.2, \quad \widehat{\text{Cl}}_3 = 0.6.$$

If Step 1 selected action 3, then **confidence** = 0.6. If Step 1 selected action 1, then **confidence** = 0.2.

The corresponding implementation

The following is the relevant calculation from `mrbsp/modules/evaluators/multi_robot.py`. The helper first builds the sampled cumulative-likelihood distribution over all actions. The later lines extract the value belonging to the Step-1 action and store it as **confidence**.

```
def get_cl_per_action(choices):
    choices_per_action = {
        i: 0
        for i in range(len(list(choices.values())[0]["rewards"]))
    }

    # Number of supporting samples for each action choice.
    for sample_data in choices.values():
        mr_choice = sample_data["opt_action"]
        choices_per_action[mr_choice] += 1

    cl_per_action = {
```

```

        action: choices_per_action[action] / len(choices)
    for action in choices_per_action
}
return cl_per_action

cls_s2 = get_cl_per_action(result["step_2"])
step_1_action = result["step_1"]["optimal_action_idx"]
cl_s2 = cls_s2.get(step_1_action, 0)
confidence = cl_s2

```

For the numerical example above, the resulting dictionary would contain

```

cls_s2 = {0: 0.2, 1: 0.2, 2: 0.0, 3: 0.6}
step_1_action = 3
confidence = cls_s2[step_1_action] # 0.6

```

The original source uses the same count-divided-by-sample-count operation; the excerpt is reformatted only to make it fit and read clearly on the page.

5 What ϵ controls

ϵ is a relaxation or tolerated-risk parameter. The probabilistic branch uses the threshold

$$\widehat{\text{Cl}}_{\bar{a}} > 1 - \epsilon.$$

Thus a smaller ϵ is stricter, while a larger ϵ is more permissive. In particular, $\epsilon = 0.7$ does *not* mean 70% confidence; it produces a support threshold of $1 - 0.7 = 0.3$.

Because the implementation uses a strict inequality, the required counts for $N = 10$ are:

ϵ	Implemented threshold	Minimum supporting samples
0.1	$\widehat{\text{Cl}}_{\bar{a}} > 0.9$	10/10
0.3	$\widehat{\text{Cl}}_{\bar{a}} > 0.7$	8/10
0.5	$\widehat{\text{Cl}}_{\bar{a}} > 0.5$	6/10
0.7	$\widehat{\text{Cl}}_{\bar{a}} > 0.3$	4/10
0.9	$\widehat{\text{Cl}}_{\bar{a}} > 0.1$	2/10

6 The complete acceptance rule

The threshold is only one branch of the implemented rule. For each of Steps 2 and 3, the implementation accepts the Step-1 action when

$$\left[\widehat{\text{Cl}}_{\bar{a}} = \max_{a \in \mathcal{A}} \widehat{\text{Cl}}_a \right] \quad \vee \quad \left[\widehat{\text{Cl}}_{\bar{a}} > 1 - \epsilon \right].$$

The robot accepts R-VerifyAC only when this is true in *both* Step 2 and Step 3:

$$\text{Accept}_r(\bar{a}) = \text{Accept}_r^{(2)}(\bar{a}) \wedge \text{Accept}_r^{(3)}(\bar{a}).$$

The first branch is the modal-action branch: the Step-1 action has the largest sampled support. The second is the $1 - \epsilon$ probabilistic branch. As a result, a Step-1 action can pass even below the numerical threshold when it is modal.

For example, if ten samples produce counts (3, 3, 2, 2) and the Step-1 action is one of the actions with three votes, the current code accepts it through the modal branch even when $\epsilon = 0.7$, because $0.3 > 0.3$ is false but 0.3 is tied for the maximum.

7 Step 2 versus Step 3

The same calculation is performed from two complementary reasoning directions:

- Step 2 asks how the reasoner’s preferred action changes across plausible missing observations of its peer.
- Step 3 mirrors that reasoning from the peer’s side.

Consequently there are two relevant support estimates per robot, $\widehat{\text{Cl}}_a^{(2)}$ and $\widehat{\text{Cl}}_a^{(3)}$. A single scalar called `conf` cannot by itself represent the complete acceptance decision.

8 Important implementation audit findings

1. **The top-level `conf` field is Step 2 only.** The main evaluator stores `conf` as the reference robot’s Step 2 value, although final acceptance depends on both Step 2 and Step 3.
2. **Ties are accepted by the code.** The implementation tests equality with the maximum sampled support. The paper defines the rank-1 action using strict dominance over every other action. Therefore a tied modal action passes in code but does not satisfy that strict unique-rank-1 definition.
3. **A legacy post-processing path uses a different rule.** It chooses the first iteration satisfying `conf >= prob_threshold`. This uses only Step 2, uses greater-than-or-equal instead of strict greater-than, and omits the explicit Step 3/modal structure. It should not be treated as the authoritative R-VerifyAC acceptance test.
4. **Finite-sample uncertainty is separate from ϵ .** ϵ sets the desired support threshold. It does not quantify how accurate $\widehat{\text{Cl}}$ is as an estimate of the true Cl . That requires a separate concentration parameter and a bound such as Hoeffding’s inequality.

9 Terminology recommended for tables and plots

Use	Meaning
Estimated cumulative likelihood	$\widehat{\text{Cl}}_a$, the sampled support of any action a
Step-2 support of Step-1 action	$\widehat{\text{Cl}}_a^{(2)}$
Step-3 support of Step-1 action	$\widehat{\text{Cl}}_a^{(3)}$
Acceptance threshold	$1 - \epsilon$
R-VerifyAC accepted	Both Step-2 and Step-3 Boolean tests passed

Avoid presenting `confidence` without specifying the step, and avoid the phrase “cumulative entropy.”

10 Code locations audited

- Per-action sample fractions and acceptance: `mrbsp/modules/evaluators/multi_robot.py`, lines 1167–1178 and 1249–1259.
- Stored top-level `conf`: the same file, line 1384.
- Experiment defaults: `scenarios/anpl_mobile_platform/scripts/multi_robot_planning.py`, lines 2643–2645.
- Legacy confidence-only iteration selection: the same file, lines 3311–3315.
- Paper definition and estimator: `continuous-and-high-dim-cases.tex`, cumulative-likelihood section.